Many-body perturbation theory: reference states, projection methods and energy corrections

FYS4480/FYS9480 – Week 44, October 29–30, 2026

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Thursday: a reference determinant and its excitations, the exact energy projected onto it, projection operators and the reduced resolvent, Brillouin-Wigner versus Rayleigh-Schrödinger, and the first three orders

Friday: the second-order energy from Wick's theorem, a two-level benchmark through fourth order, and every contribution to the energy through fourth order and to the wave operator through third

Reading: lecture notes chapter 9, sections 9.1–9.6; Shavitt and Bartlett, chapter 2; Szabo and Ostlund, chapter 6. Code: `BookPrograms/chapter09/mbpt.py`; notebook `week44.ipynb`.

Exercises: Last two slides; answers on the week-45 slides.

Plan for week 44

Thursday, 2×45 min: reference states, projection methods and systematic orders (sections 9.1–9.4)

- A reference determinant, its $1p1h$ and $2p2h$ excitations, and the exact energy projected onto the reference: only singles and doubles couple directly
- A two-state problem solved by hand: the orders we must recover
- Intermediate normalisation, two energy differences that must be kept apart, P and Q, the reduced resolvent, an exact equation for $Q|\Psi_0\rangle$ and its Neumann series
- Two choices of starting energy: Brillouin-Wigner ($\omega = E_0$, implicit) and Rayleigh-Schrödinger ($\omega = \varepsilon_0$, explicit)
- A resolvent identity, the resummation in Q space, the energy-dependent effective Hamiltonian, and the secular equation recovered
- Rayleigh-Schrödinger order by order: $E^{(1)}$, $E^{(2)}$, and the normalisation subtraction in $E^{(3)}$

Friday, 2×45 min: the second-order energy and a fourth-order check (sections 9.5–9.6)

- First-order direct and exchange terms; which intermediate determinants contribute; the perturbation normal-ordered, $H_I = E^{(1)} + U_N + V_N$
- Singles: the coherent mean-field sum. Doubles: Wick contractions, the counting factor, the leading amplitude. The complete $E^{(2)}$
- A purely off-diagonal two-level benchmark: Brillouin-Wigner exact with two vertices, Rayleigh-Schrödinger through fourth order, only the folded term survives
- The list: every contribution to the energy through fourth order and to the wave operator through third order

Where we are, and the route through this week

Done

- Full CI (weeks 37–38): exact, exponential
- Hartree-Fock (weeks 38–39): the best single determinant, $f_{ai} = 0$
- Stability, DMRG, the electron gas, density functional theory (weeks 39–43)

This week

1. Reference determinants and perturbative amplitudes
2. Projection operators and the reduced resolvent
3. Brillouin-Wigner and Rayleigh-Schrödinger expansions
4. Second-order energy from single and double excitations
5. A two-level check through fourth order

Many-body connection

Scope: nondegenerate, time-independent perturbation theory, with a single reference determinant and a Hamiltonian containing at most two-body terms. Next week the same terms are drawn as diagrams, and the rules for reading them off are derived.

Thursday, lecture 1: Reference states and expansions

A reference determinant and its excitations (section 9.1)

Choose an orthonormal determinant basis built on $|\Phi_0\rangle$:

$$|\Psi_0\rangle = |\Phi_0\rangle + \sum_{ia} c_i^a |\Phi_i^a\rangle + \sum_{\substack{i<j \\ a<b}} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle + \cdots, \qquad |\Phi_i^a\rangle = a_a^\dagger a_i |\Phi_0\rangle, \qquad |\Phi_{ij}^{ab}\rangle = a_a^\dagger a_b^\dagger a_j a_i |\Phi_0\rangle.$$

- $i, j, k, \ldots$: occupied orbitals; $a, b, c, \ldots$: unoccupied orbitals (weeks 37–38).
- The exact expansion reaches all allowed *NpNh* excitations.
- MBPT approximates the amplitudes order by order.

Here the reference coefficient is one; doubles are summed once, with $i < j$ and $a < b$.

Projecting the energy onto the reference

With $\langle\Phi_0|\Psi_0\rangle = 1$ and $H|\Psi_0\rangle = E_0|\Psi_0\rangle$,

$$E_0 = \langle\Phi_0|H|\Psi_0\rangle, \qquad E_{\mathrm{ref}} = \langle\Phi_0|H|\Phi_0\rangle.$$

For an at-most-two-body Hamiltonian,

$$\boxed{E_0 - E_{\mathrm{ref}} = \sum_{ia} c_i^a f_{ia} + \sum_{\substack{i<j \\ a<b}} c_{ij}^{ab} \overline{v}_{ijab}.} \qquad f_{ia} \equiv \langle\Phi_0|H|\Phi_i^a\rangle, \quad \overline{v}_{ijab} \equiv \langle ij|v|ab\rangle_{\mathrm{AS}}.$$

- Only singles and doubles couple directly to the reference (the energy equation of week 38).
- Higher excitations still influence the exact amplitudes.

Preview: leading amplitudes and second-order energy

For $H_0 = \sum_p \varepsilon_p a_p^\dagger a_p$, with $\varepsilon_0 = \sum_{i \in \mathrm{occ}} \varepsilon_i$, the leading amplitudes will turn out to be

$$c_i^{a(1)} = \frac{f_{ai}}{\varepsilon_i - \varepsilon_a}, \qquad c_{ij}^{ab(1)} = \frac{\overline{v}_{abij}}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}.$$

Substitution into the projected energy gives

$$E^{(2)} = \sum_{ia} \frac{|f_{ai}|^2}{\varepsilon_i - \varepsilon_a} + \sum_{\substack{i<j \\ a<b}} \frac{|\overline{v}_{abij}|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}.$$

Hartree-Fock reference

For canonical Hartree-Fock orbitals, $f_{ai} = 0$ (Brillouin, week 38): the singles term vanishes. This is a special case, not a condition on a general reference.

A two-state illustration of the perturbative idea

Use a one-particle model to make the algebra transparent:

$$H = H_0 + H_I, \qquad H_0 = \sum_{p=1}^{2} \varepsilon_p a_p^\dagger a_p, \qquad H_I = g \sum_{p,q=1}^{2} a_p^\dagger a_q,$$

with $\varepsilon_1 < \varepsilon_2$ and $|p\rangle = a_p^\dagger |0\rangle$.

$$\langle p|q\rangle = \delta_{pq}, \qquad P = |1\rangle\langle 1|, \qquad Q = |2\rangle\langle 2|, \qquad P + Q = 1.$$

For example,

$$\langle 1|H_I|2\rangle = g \sum_{pq} \langle 0|a_1 a_p^\dagger a_q a_2^\dagger|0\rangle = g \sum_{pq} \delta_{1p}\delta_{q2} = g.$$

This is an algebraic benchmark inside the many-body development; it is not an interacting many-particle system.

Exact eigenvalues of the two-state model

In the basis $\{|1\rangle, |2\rangle\}$,

$$H = \begin{pmatrix} \varepsilon_1 + g & g \\ g & \varepsilon_2 + g \end{pmatrix}, \qquad \det(H - EI) = 0, \qquad E_\pm = \frac{\varepsilon_1 + \varepsilon_2}{2} + g \pm \sqrt{\frac{(\varepsilon_2 - \varepsilon_1)^2}{4} + g^2}.$$

Define the positive gap $\Delta = \varepsilon_2 - \varepsilon_1 > 0$. Then

$$\boxed{E_- = \varepsilon_1 + g + \frac{\Delta}{2}\left[1 - \sqrt{1 + \frac{4g^2}{\Delta^2}}\right].}$$

The positive gap fixes the branch and the signs before expanding the square root.

Reading off perturbative orders

Use $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots$ with $x = 4g^2/\Delta^2$. The lower eigenvalue becomes

$$\boxed{E_- = \varepsilon_1 + g - \frac{g^2}{\Delta} + \frac{g^4}{\Delta^3} + \mathcal{O}(g^6).}$$

Unperturbed	First order	Second order	Third order	Fourth order
ε_1	g	$-g^2/\Delta$	0	g^4/Δ^3

The task of perturbation theory

Recover these contributions from the Hamiltonian and the reference state, without solving the full eigenvalue problem first.

Formal MBPT and intermediate normalisation (section 9.1)

$$H = H_0 + H_I, \qquad H_0|\Phi_m\rangle = \varepsilon_m|\Phi_m\rangle, \qquad H|\Psi_0\rangle = E_0|\Psi_0\rangle.$$

Assume a nondegenerate reference $|\Phi_0\rangle$ with nonzero overlap with the exact state.

$$\langle\Phi_0|\Phi_0\rangle = 1, \qquad \boxed{\langle\Phi_0|\Psi_0\rangle = 1}, \qquad |\Psi_0\rangle = |\Phi_0\rangle + \sum_{m\neq 0} c_m|\Phi_m\rangle.$$

- The reference coefficient is fixed to one.
- Intermediate normalisation does not require $\langle\Psi_0|\Psi_0\rangle = 1$.
- All excited determinants are included in the label $m \neq 0$.

Pause to think #1

(about 3 min – discuss with your neighbour)

Intermediate normalisation sets $\langle \Phi_0 | \Psi_0 \rangle = 1$ instead of $\langle \Psi_0 | \Psi_0 \rangle = 1$.

1. Why is this a convenience and not an approximation?
2. Which assumption about $|\Phi_0\rangle$ is needed for it to be possible at all?
3. The exact energy is $E_0 = \langle \Phi_0 | H | \Psi_0 \rangle$. Is this an expectation value? Is it an upper bound?

Answer to pause #1

1. The Schrödinger equation is linear: any multiple of $|\Psi_0\rangle$ is a solution with the same energy. Choosing the multiple such that the reference coefficient is one changes the norm, not the physics; the normalised state is $|\Psi_0\rangle/\sqrt{\langle\Psi_0|\Psi_0\rangle}$.
2. $\langle\Phi_0|\Psi_0\rangle \neq 0$: the reference must have a non-zero component along the exact ground state. If it does not – wrong symmetry, wrong particle number, or a reference that is orthogonal to the target – no rescaling can produce the coefficient one.
3. It is a *transition* matrix element, not an expectation value $\langle\Psi_0|H|\Psi_0\rangle/\langle\Psi_0|\Psi_0\rangle$. The two agree only for the exact state; for an approximate $|\Psi_0\rangle$ the projected energy is *not* variational and can lie below E_0. This is why perturbation theory can cross the exact answer from either side.

Two energy differences that must be distinguished

Reference projection gives

$$E_0 = \varepsilon_0 + \langle \Phi_0 | H_I | \Psi_0 \rangle, \qquad E^{(1)} = \langle \Phi_0 | H_I | \Phi_0 \rangle, \qquad E_{\mathrm{ref}} = \langle \Phi_0 | H | \Phi_0 \rangle = \varepsilon_0 + E^{(1)}.$$

Energy shift relative to the unperturbed state

$$\Delta E \equiv E_0 - \varepsilon_0 = E^{(1)} + E^{(2)} + \cdots .$$

Correlation energy relative to the reference determinant

$$E_0 - E_{\mathrm{ref}} = E^{(2)} + E^{(3)} + \cdots .$$

The first-order term is the mean-field energy of the reference, not correlation; whether it vanishes depends on the partition. Correlation begins at second order.

Separating the reference and its complement (section 9.2)

$$P = |\Phi_0\rangle\langle\Phi_0|, \qquad Q = \sum_{m\neq 0} |\Phi_m\rangle\langle\Phi_m|,$$

$$P^2 = P, \quad Q^2 = Q, \quad P + Q = 1, \quad PQ = QP = 0, \qquad [P, H_0] = [Q, H_0] = 0.$$

Intermediate normalisation implies

$$P|\Psi_0\rangle = |\Phi_0\rangle, \qquad \boxed{|\Psi_0\rangle = |\Phi_0\rangle + Q|\Psi_0\rangle.}$$

- P: one-dimensional model space.
- Q: all configurations excited from the reference.

The reduced resolvent

For a scalar starting energy ω, define

$$\boxed{R_Q(\omega) \equiv Q(\omega - H_0)^{-1} Q = \sum_{m \neq 0} \frac{|\Phi_m\rangle\langle\Phi_m|}{\omega - \varepsilon_m}.}$$

$$R_Q(\omega)|\Phi_0\rangle = 0, \qquad R_Q(\omega)|\Phi_m\rangle = \frac{|\Phi_m\rangle}{\omega - \varepsilon_m} \quad (m \neq 0).$$

The inverse is restricted to Q space

At $\omega = \varepsilon_0$, the full inverse has a reference-state pole. The reduced resolvent remains defined if $\varepsilon_0 \neq \varepsilon_m$ for all $m \neq 0$ – this is where non-degeneracy is used.

An exact equation for the correlated component

Rewrite the Schrödinger equation as

$$(\omega - H_0)|\Psi_0\rangle = (\omega - E_0 + H_I)|\Psi_0\rangle.$$

Project and apply the reduced inverse:

$$Q|\Psi_0\rangle = R_Q(\omega)(\omega - E_0 + H_I)|\Psi_0\rangle, \qquad \boxed{|\Psi_0\rangle = |\Phi_0\rangle + R_Q(\omega)(\omega - E_0 + H_I)|\Psi_0\rangle.}$$

- The Hamiltonian and reference are known.
- The exact energy E_0 and correlated state are still unknown: exact, and useless as it stands.

Iterating the wave-function equation

Set $A(\omega) = R_Q(\omega)(\omega - E_0 + H_I)$. Repeated substitution yields

$$|\Psi_0\rangle = |\Phi_0\rangle + A|\Psi_0\rangle = |\Phi_0\rangle + A|\Phi_0\rangle + A^2|\Psi_0\rangle = |\Phi_0\rangle + A|\Phi_0\rangle + A^2|\Phi_0\rangle + \cdots .$$

Formally,

$$\boxed{|\Psi_0\rangle = \sum_{n=0}^{\infty} \left[R_Q(\omega)(\omega - E_0 + H_I)\right]^n |\Phi_0\rangle,} \qquad \Delta E = \sum_{n=0}^{\infty} \langle\Phi_0| H_I \left[R_Q(\omega)(\omega - E_0 + H_I)\right]^n |\Phi_0\rangle.$$

A trap to avoid

This is a formal Neumann series. Its convergence is not guaranteed by the algebraic iteration alone; Friday's benchmark shows the series diverging while the exact answer stays perfectly smooth.

Two choices of starting energy (section 9.3)

Brillouin-Wigner (BW)

$$\omega = E_0.$$

The unknown exact energy enters the denominators; the equation is implicit and must be solved self-consistently.

Rayleigh-Schrödinger (RS)

$$\omega = \varepsilon_0.$$

Fixed unperturbed denominators; collect equal powers of the interaction, and expand ΔE in the numerators too.

The first terms of the Brillouin-Wigner series, with $\langle X \rangle_0 \equiv \langle \Phi_0 | X | \Phi_0 \rangle$ and $R_E = R_Q(E_0)$:

$$\Delta E = \langle H_I \rangle_0 + \langle H_I R_E H_I \rangle_0 + \langle H_I R_E H_I R_E H_I \rangle_0 + \cdots, \qquad R_E = \sum_{m \neq 0} \frac{|\Phi_m\rangle\langle\Phi_m|}{E_0 - \varepsilon_m}.$$

- Terms are ordered by the number of explicit interaction vertices.
- Because E_0 itself depends on the interaction, a finite BW truncation is *not* a Taylor expansion in the coupling.

Thursday, lecture 2: Projection methods and orders

The projection framework in compact form

$$P = |\Phi_0\rangle\langle\Phi_0|, \qquad Q = 1 - P, \qquad R_Q(\omega) = \sum_{m\neq 0} \frac{|\Phi_m\rangle\langle\Phi_m|}{\omega - \varepsilon_m},$$

$$\Delta E = \sum_{n=0}^{\infty} \langle H_I \left[R_Q(\omega)(\omega - E_0 + H_I)\right]^n \rangle_0.$$

Keep the two baselines explicit:

$$\Delta E = E_0 - \varepsilon_0 = \langle\Phi_0|H_I|\Psi_0\rangle, \qquad E_0 - E_{\rm ref} = \sum_{m\neq 0} c_m \langle\Phi_0|H_I|\Phi_m\rangle, \qquad E_{\rm ref} = \varepsilon_0 + \langle\Phi_0|H_I|\Phi_0\rangle.$$

For $\omega = E_0$: $\Delta E = \langle H_I\rangle_0 + \langle H_I R_E H_I\rangle_0 + \langle H_I R_E H_I R_E H_I\rangle_0 + \cdots$.

Next step

Sum the repeated propagation and interactions within Q space into one energy-dependent resolvent.

A resolvent identity

For invertible operators A and $A - B$,

$$\boxed{(A-B)^{-1} = A^{-1} + A^{-1}B(A-B)^{-1}.}$$

Check by multiplying on the right by $A - B$:

$$\left[A^{-1} + A^{-1}B(A-B)^{-1}\right](A-B) = 1 - A^{-1}B + A^{-1}B = 1.$$

Iterating gives

$$(A-B)^{-1} = A^{-1} + A^{-1}BA^{-1} + A^{-1}BA^{-1}BA^{-1} + \cdots .$$

Preserve the order of the operator factors; A and B need not commute.

Resumming propagation in the complementary space

Take $A = E_0 - QH_0Q$ and $B = QH_IQ$, acting within Q space. Define

$$G_Q(E_0) = Q\,(E_0 - QHQ)^{-1}_Q\,Q = Q\,(E_0 - H_0 - QH_IQ)^{-1}_Q\,Q.$$

Its formal expansion is

$$G_Q(E_0) = R_E + R_EH_IR_E + R_EH_IR_EH_IR_E + \cdots,$$

where every intermediate propagation is restricted to Q; the subscript Q specifies where the inverse is taken.

Many-body connection

G_Q is the resolvent of the *full* Hamiltonian restricted to the excited space. It is the object a Green's function approach (chapter 12) would start from.

An exact energy-dependent effective Hamiltonian

The projected Schrödinger equation gives $Q|\Psi_0\rangle = G_Q(E_0)H_I|\Phi_0\rangle$. Hence

$$\boxed{\Delta E = \langle H_I\rangle_0 + \langle H_I G_Q(E_0) H_I\rangle_0,} \qquad \boxed{|\Psi_0\rangle = [1 + G_Q(E_0)H_I]\,|\Phi_0\rangle.}$$

Equivalently, in the model space,

$$H_{\rm eff}(E_0) = PHP + PHQ(E_0 - QHQ)^{-1}_Q QHP.$$

These equations are exact where the reference overlap is nonzero and the required Q-space inverse exists. Their energy dependence makes them implicit (Feshbach–Löwdin partitioning).

Applying the projected resolvent to two states

Return to

$$H = \begin{pmatrix} \varepsilon_1 + g & g \\ g & \varepsilon_2 + g \end{pmatrix}, \qquad P = |1\rangle\langle 1|, \quad Q = |2\rangle\langle 2|.$$

There is only one state in Q:

$$QHQ = (\varepsilon_2 + g)Q, \qquad G_Q(E_0) = \frac{|2\rangle\langle 2|}{E_0 - \varepsilon_2 - g}, \qquad \langle 1|H_I|1\rangle = g, \qquad \langle 1|H_I|2\rangle = \langle 2|H_I|1\rangle = g.$$

The resummed BW equation becomes

$$E_0 - \varepsilon_1 = g + \frac{g^2}{E_0 - \varepsilon_2 - g} \quad \Longleftrightarrow \quad \boxed{(E_0 - \varepsilon_1 - g)(E_0 - \varepsilon_2 - g) - g^2 = 0,}$$

which is precisely $\det(H - E_0) = 0$.

What is exact here?

The complete Q-space resummation plus the self-consistent solution reproduces the exact eigenvalues. This is not a claim that ordinary second-order perturbation theory is exact.

Rayleigh-Schrödinger: fix the starting energy (section 9.4)

Set $\omega = \varepsilon_0$ and introduce

$$R_0 = \frac{Q}{\varepsilon_0 - H_0}, \qquad \Delta E = E_0 - \varepsilon_0.$$

Then

$$|\Psi_0\rangle = |\Phi_0\rangle + R_0(H_I - \Delta E)|\Psi_0\rangle, \qquad \boxed{\Delta E = \sum_{n=0}^{\infty} \langle H_I \left[R_0(H_I - \Delta E)\right]^n \rangle_0.}$$

- The denominators now involve unperturbed energies only.
- The energy shift inside the operator product must itself be expanded.
- $R_0|\Phi_0\rangle = 0$ removes a rightmost scalar subtraction.

Collect terms of the same perturbative order

Introduce a bookkeeping parameter η:

$$H(\eta) = H_0 + \eta H_I, \qquad E_0(\eta) = \varepsilon_0 + \sum_{n\geq 1} \eta^n E^{(n)}.$$

The first terms of the implicit energy equation are

$$\Delta E = \langle H_I \rangle_0 + \langle H_I R_0 (H_I - \Delta E) \rangle_0 + \langle H_I R_0 (H_I - \Delta E) R_0 H_I \rangle_0 + \cdots .$$

Since $R_0|\Phi_0\rangle = 0$, $\langle H_I R_0 (H_I - \Delta E) \rangle_0 = \langle H_I R_0 H_I \rangle_0$.

A trap to avoid

Set $\eta = 1$ after collecting equal powers. A superscript (n) denotes an energy contribution of order n in the interaction, *not* an excitation rank: a first-order amplitude will feed the second-order energy.

The first- and second-order corrections

$$\boxed{E^{(1)} = \langle H_I \rangle_0, \qquad E^{(2)} = \langle H_I R_0 H_I \rangle_0.}$$

Insert a resolution over excited determinants:

$$\boxed{E^{(2)} = \sum_{m \neq 0} \frac{\langle \Phi_0 | H_I | \Phi_m \rangle \langle \Phi_m | H_I | \Phi_0 \rangle}{\varepsilon_0 - \varepsilon_m}} \qquad \text{(Hermitian } H_I\text{)} \qquad E^{(2)} = \sum_{m \neq 0} \frac{|\langle \Phi_m | H_I | \Phi_0 \rangle|^2}{\varepsilon_0 - \varepsilon_m}.$$

For a nondegenerate lowest unperturbed state, every denominator is negative: the second-order energy is nonpositive.

Third order includes a normalisation subtraction

Collecting terms cubic in H_I,

$$\boxed{E^{(3)} = \langle H_I R_0 H_I R_0 H_I \rangle_0 - E^{(1)} \langle H_I R_0^2 H_I \rangle_0.}$$

Writing $V_{mn} = \langle \Phi_m | H_I | \Phi_n \rangle$,

$$E^{(3)} = \sum_{m,n\neq 0} \frac{V_{0m} V_{mn} V_{n0}}{(\varepsilon_0 - \varepsilon_m)(\varepsilon_0 - \varepsilon_n)} - V_{00} \sum_{m\neq 0} \frac{V_{0m} V_{m0}}{(\varepsilon_0 - \varepsilon_m)^2}.$$

- The subtraction contains $E^{(1)}$, not the full ΔE.
- It comes from regrouping the energy-dependent subtraction in the RS equation.

Measured

Pairing model, four levels, two pairs, $g = 0.5$: principal term -0.02300347, subtraction -0.01258681, $E^{(3)} = -0.01041667$. The "correction" is larger than the answer.

Pause to think #2

(about 4 min – discuss with your neighbour)

For the two-state model $H_I = g\sum_{pq} a_p^\dagger a_q$, the exact expansion has *no* third-order term.

1. Compute the principal third-order term $\langle H_I R_0 H_I R_0 H_I\rangle_0$ and the subtraction $E^{(1)}\langle H_I R_0^2 H_I\rangle_0$ with $R_0 = |2\rangle\langle 2|/D$, $D = \varepsilon_1 - \varepsilon_2$.
2. Why does the third order vanish? Would it vanish if $\langle 2|H_I|2\rangle \neq \langle 1|H_I|1\rangle$?

Answer to pause #2

1. Principal: $V_{12}V_{22}V_{21}/D^2 = g \cdot g \cdot g/D^2 = g^3/D^2$. Subtraction: $V_{11}V_{12}V_{21}/D^2 = g^3/D^2$. Difference zero.
2. It vanishes because the diagonal interaction is the same in the model space and in Q: $V_{22} = V_{11}$. In general $E^{(3)} = V_{12}^2(V_{22} - V_{11})/D^2$, and for $V_{22} \neq V_{11}$ it does not vanish. The cancellation is between a "real" term and the normalisation subtraction – the subtraction is not a small correction.

The total energy through second order

$$\boxed{E_0^{[2]} = \varepsilon_0 + E^{(1)} + E^{(2)} = E_{\rm ref} + E^{(2)}.}$$

For the partition $H = \sum_{pq} h^0_{pq} a^\dagger_p a_q + V$, with $H_I = V$ a bare two-body interaction,

$$V = \frac{1}{4}\sum_{pqrs} \overline{v}_{pqrs} a^\dagger_p a^\dagger_q a_s a_r, \qquad \boxed{E^{(1)} = \frac{1}{2}\sum_{ij\in \rm occ} \overline{v}_{ijij}.}$$

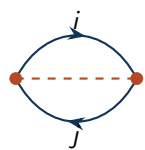
The form of $E^{(1)}$ depends on the partition. If an auxiliary mean field is included in H_0, its subtraction belongs in $H_I = H - H_0$.

First-order direct and exchange contributions

$$E^{(1)} = \frac{1}{2} \sum_{ij \in \mathrm{occ}} \left[\langle ij|v|ij\rangle - \langle ij|v|ji\rangle \right].$$



Direct: $+\frac{1}{2}\sum_{ij}\langle ij|v|ij\rangle$.



Exchange: $-\frac{1}{2}\sum_{ij}\langle ij|v|ji\rangle$.

Dashed line: two-body interaction. Solid loops: occupied-line contractions. The exchange sign is displayed explicitly in the algebra. Next week: the rules that produce the sign and the $\frac{1}{2}$ from the drawing.

Which intermediate determinants contribute?

$$E^{(2)} = \sum_{m \neq 0} \frac{\langle \Phi_0 | H_I | \Phi_m \rangle \langle \Phi_m | H_I | \Phi_0 \rangle}{\varepsilon_0 - \varepsilon_m}.$$

For an interaction containing at most two-body operators,

$$|\Phi_0\rangle \xrightarrow{H_I} \left\{ |\Phi_i^a\rangle,\ |\Phi_{ij}^{ab}\rangle \right\} \xrightarrow{H_I} |\Phi_0\rangle.$$

Two sectors at second order

$$E^{(2)} = E^{(2)}_{1p1h} + E^{(2)}_{2p2h}.$$

A reference component is excluded by Q. No triple or higher excitation can be reached from a determinant by a single two-body vertex.

Normal ordering exposes the mean-field contraction

For the partition just introduced (week 37, section 3.17),

$$H = E_{\mathrm{ref}} + F_N + V_N, \qquad F_N = \sum_{pq} (h^0_{pq} + U_{pq}) \{ a^\dagger_p a_q \},$$

$$V_N = \frac{1}{4} \sum_{pqrs} \overline{v}_{pqrs} \{ a^\dagger_p a^\dagger_q a_s a_r \}, \qquad \boxed{U_{pq} = \sum_{j \in \mathrm{occ}} \overline{v}_{pjqj}.}$$

- Braces denote normal ordering with respect to $|\Phi_0\rangle$.
- U is the mean field called u^{HF} in the notes.
- This definition does not assume that the orbitals solve the HF equations.

The interaction in normal-ordered form

Since $H_0 = \sum_{pq} h^0_{pq} a^\dagger_p a_q$ here,

$$\boxed{H_I = V = E^{(1)} + U_N + V_N,} \qquad U_N = \sum_{pq} U_{pq} \{ a^\dagger_p a_q \}, \qquad V_N = \frac{1}{4} \sum_{pqrs} \overline{v}_{pqrs} \{ a^\dagger_p a^\dagger_q a_s a_r \}.$$

- The scalar $E^{(1)}$ cannot connect P and Q.
- U_N connects the reference to singles.
- V_N connects the reference to doubles.

A trap to avoid

The braces matter: adding U_N to an un-normal-ordered bare two-body operator would count the mean-field contractions twice.

Evaluating the singles matrix element

For $|\Phi_i^a\rangle = a_a^\dagger a_i|\Phi_0\rangle$,

$$\langle\Phi_i^a|a_p^\dagger a_q|\Phi_0\rangle = \langle\Phi_0|a_i^\dagger a_a a_p^\dagger a_q|\Phi_0\rangle = \delta_{pa}\delta_{qi}.$$

The two nonzero contractions pair the occupied operators and the unoccupied operators:

$$(a_i^\dagger, a_q) \longrightarrow \delta_{iq}, \qquad (a_a, a_p^\dagger) \longrightarrow \delta_{ap}.$$

Therefore

$$\boxed{\langle\Phi_i^a|H_I|\Phi_0\rangle = U_{ai}, \qquad \langle\Phi_0|H_I|\Phi_i^a\rangle = U_{ia} = U_{ai}^*.}$$

Compare week 39: this is the Hartree-Fock matrix element $\langle a|f|i\rangle$ with h_0 diagonal.

The singles contribution to the second-order energy

The intermediate determinant has unperturbed energy $\varepsilon_i^a = \varepsilon_0 - \varepsilon_i + \varepsilon_a$. Consequently,

$$\boxed{E_{1p1h}^{(2)} = \sum_{ia} \frac{U_{ia} U_{ai}}{\varepsilon_i - \varepsilon_a} = \sum_{ia} \frac{|U_{ai}|^2}{\varepsilon_i - \varepsilon_a}.}$$

Since $U_{ai} = \sum_{j \in \mathrm{occ}} \overline{v}_{ajij}$,

$$E_{1p1h}^{(2)} = \sum_{ia} \frac{\left| \sum_{j \in \mathrm{occ}} \overline{v}_{ajij} \right|^2}{\varepsilon_i - \varepsilon_a}.$$

The occupied-orbital contributions add before squaring. In general, $\left| \sum_j z_j \right|^2 \neq \sum_j |z_j|^2$.

The leading singles amplitude and the CI energy

When h_0 is diagonal,

$$\langle i|h_0|a\rangle = \varepsilon_a\langle i|a\rangle = 0, \qquad f_{ia} = h^0_{ia} + U_{ia} = U_{ia}.$$

The exact reference-projected singles energy is $(E_0 - E_{\mathrm{ref}})_{\mathrm{singles}} = \sum_{ia} c_i^a U_{ia}$. Its leading perturbative approximation uses

$$\boxed{c_i^{a(1)} = \frac{U_{ai}}{\varepsilon_i - \varepsilon_a},} \qquad \sum_{ia} c_i^{a(1)} U_{ia} = \sum_{ia} \frac{|U_{ai}|^2}{\varepsilon_i - \varepsilon_a}.$$

Many-body connection

The exact amplitude c_i^a comes from the projected FCI equations of week 38; the perturbative amplitude is its leading term, and the second-order energy is the projected energy evaluated with it.

Friday, lecture 3: Second order and a fourth-order check

Connecting the CI energy and the perturbative expansion (section 9.5)

For exact CI amplitudes,

$$E_0 - E_{\mathrm{ref}} = \sum_{ia} c_i^a f_{ia} + \frac{1}{4} \sum_{ijab} c_{ij}^{ab} \overline{v}_{ijab}.$$

Here and below, i, j, a, b are unrestricted within their occupied or unoccupied sets, and the doubles amplitudes are antisymmetric.

$$E_0 - \varepsilon_0 = E^{(1)} + E^{(2)} + \cdots, \qquad E^{(2)} = \sum_{m \neq 0} \frac{\langle \Phi_0 | H_I | \Phi_m \rangle \langle \Phi_m | H_I | \Phi_0 \rangle}{\varepsilon_0 - \varepsilon_m}.$$

The perturbative amplitudes approximate the exact configuration amplitudes. The energy baseline remains $E_{\mathrm{ref}} = \varepsilon_0 + E^{(1)}$.

Singles: the coherent mean-field sum

For the diagonal-h_0, bare-interaction partition,

$$f_{ia} = U_{ia}, \qquad U_{ai} = \sum_{j\in\mathrm{occ}} \overline{v}_{ajij}, \qquad c_i^{a(1)} = \frac{U_{ai}}{\varepsilon_i - \varepsilon_a}.$$

Then

$$E^{(2)}_{1p1h} = \sum_{ia} \frac{U_{ia}U_{ai}}{\varepsilon_i - \varepsilon_a} = \sum_{ia}\sum_{jk\in\mathrm{occ}} \frac{\overline{v}_{ijaj}\overline{v}_{akik}}{\varepsilon_i - \varepsilon_a}.$$

Keep the cross terms

The j and k sums are independent: this is the modulus squared of a sum over occupied orbitals, not the sum of individual moduli squared.

Pause to think #3

(about 3 min – discuss with your neighbour)

The singles term $E^{(2)}_{1p1h} = \sum_{ia} |\sum_j v_{ajij}|^2/(\varepsilon_i - \varepsilon_a)$ is a double sum over occupied j, k.

1. For the pairing model (levels p, spins $\pm$, interaction moving pairs only), what is U_{ai}? What is $E^{(2)}_{1p1h}$?
2. For the pairing plus particle-hole model of week 39 the answer is different. Why?
3. What happens to this term in a Hartree-Fock basis – and what does the mean field have to be for that to happen?

Answer to pause #3

1. The pairing interaction $\overline{v}_{ajij}$ needs a and i to be the two members of the same pair, which is impossible for a empty and i occupied in the reference: $U_{ai} = 0$, and the singles term vanishes identically, for every g and every basis.
2. The particle-hole term breaks pairs and connects $|\Phi_0\rangle$ to $1p1h$ states, so $U_{ai} \neq 0$; its value is a coherent sum over the two occupied levels, and the squared sum is what the week-45 table will list as $E^{(2)}_{1p1h}$.
3. $f_{ai} = 0$ by Brillouin's theorem: the HF mean field u^{HF} is exactly the U defined here, evaluated in the self-consistent orbitals, so the whole $1p1h$ sector disappears at second order – which is the practical reason to build perturbation theory on a Hartree-Fock reference.

Doubles: two interaction vertices before contraction

For a double-excitation intermediate state, use

$$A_{pqrs} = a_p^\dagger a_q^\dagger a_s a_r, \qquad V = \frac{1}{4}\sum_{pqrs} \overline{v}_{pqrs} A_{pqrs}.$$

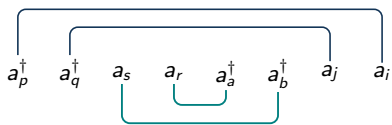
The doubles sector is

$$E^{(2)}_{2p2h} = \frac{1}{16} \sum_{m \in 2p2h} \sum_{pqrs} \sum_{tuvw} \frac{\overline{v}_{pqrs}\overline{v}_{tuvw}}{\varepsilon_0 - \varepsilon_m} \langle \Phi_0 | A_{pqrs} | \Phi_m \rangle \langle \Phi_m | A_{tuvw} | \Phi_0 \rangle.$$

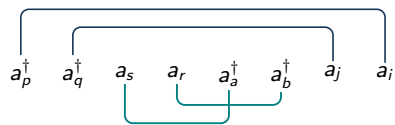
- Each vertex contributes a factor $1/4$, hence $1/16$ before contractions.
- In this formula, each intermediate determinant m is counted once.

Wick contractions and antisymmetry

Evaluate $\langle\Phi_0|a_p^\dagger a_q^\dagger a_s a_r a_a^\dagger a_b^\dagger a_j a_i|\Phi_0\rangle$:



$$+\delta_{pi}\delta_{qj}\delta_{ra}\delta_{sb} \quad \Rightarrow \quad +\overline{v}_{ijab}.$$



$$-\delta_{pi}\delta_{qj}\delta_{rb}\delta_{sa} \quad \Rightarrow \quad -\overline{v}_{ijba} = +\overline{v}_{ijab}.$$

Exchanging the occupied contractions gives two more equivalent terms. Together, four contractions cancel the $1/4$ at a vertex:

$$\boxed{\langle\Phi_0|V|\Phi_{ij}^{ab}\rangle = \overline{v}_{ijab}.}$$

The doubles denominator and the counting factor

The unperturbed double excitation has energy $\varepsilon_{ij}^{ab} = \varepsilon_0 + \varepsilon_a + \varepsilon_b - \varepsilon_i - \varepsilon_j$. An unrestricted index sum counts each determinant four times:

$$Q_{2p2h} = \sum_{\substack{i<j \\ a<b}} |\Phi_{ij}^{ab}\rangle\langle\Phi_{ij}^{ab}| = \frac{1}{4}\sum_{ijab} |\Phi_{ij}^{ab}\rangle\langle\Phi_{ij}^{ab}|.$$

Therefore

$$\boxed{E_{2p2h}^{(2)} = \frac{1}{4}\sum_{ijab} \frac{\overline{v}_{ijab}\overline{v}_{abij}}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}.}$$

For Hermitian V, the numerator is $|\overline{v}_{abij}|^2$. With $i < j$ and $a < b$, omit the factor $1/4$.

Pause to think #4

(about 3 min – discuss with your neighbour)

Factors of two and four.

1. The vertex carries $\frac{1}{4}$ and the four Wick contractions cancel it. The sum over intermediate determinants carries another $\frac{1}{4}$ if the indices are unrestricted. Which $\frac{1}{4}$ survives in $E^{(2)}_{2p2h}$, and why?
2. For the pairing model with four levels and two pairs, how many double excitations are there, and what is $E^{(2)}$ in terms of g? (Week 38: the first iterate of the FCI equations.)

Answer to pause #4

1. The vertex $\frac{1}{4}$ is cancelled by the four ways of contracting $a_p^\dagger a_q^\dagger a_s a_r$ with the excitation, which all give the same antisymmetrised element – that is what antisymmetrised elements are for. The counting $\frac{1}{4}$ survives only if one *chooses* to sum over all i, j, a, b; with $i < j$, $a < b$ each determinant appears once and there is no factor. The two conventions give the same number; mixing them gives a factor four.
2. Four: each of the two occupied pairs lifted to each of the two empty levels, matrix element $-g/2$, denominator $2(\varepsilon_i - \varepsilon_a) \in \{-2, -4, -4, -6\}$ for unit spacing: $E^{(2)} = \frac{g^2}{4}\left(-\frac{1}{2} - \frac{1}{4} - \frac{1}{4} - \frac{1}{6}\right) = -\frac{7}{24}g^2$, the first iterate of the projected equations of week 38.

The leading doubles amplitude

The first-order state correction contains $|\Psi_0^{(1)}\rangle = R_0 H_I |\Phi_0\rangle$. Projecting onto a double excitation gives

$$\boxed{c_{ij}^{ab(1)} = \frac{\langle \Phi_{ij}^{ab} | H_I | \Phi_0 \rangle}{\varepsilon_0 - \varepsilon_{ij}^{ab}} = \frac{\overline{v}_{abij}}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}.}$$

The corresponding CI energy contribution becomes

$$\frac{1}{4} \sum_{ijab} c_{ij}^{ab(1)} \overline{v}_{ijab} = E_{2p2h}^{(2)}.$$

A first-order wave-function amplitude enters the second-order energy. Excitation rank and perturbative order are different labels.

The complete second-order energy

For the bare-interaction partition with diagonal h_0,

$$E^{(2)} = \sum_{ia} \frac{|U_{ai}|^2}{\varepsilon_i - \varepsilon_a} + \frac{1}{4} \sum_{ijab} \frac{|\overline{v}_{abij}|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b},$$

$$E_0^{[2]} = E_{\mathrm{ref}} + E^{(2)}, \qquad U_{ai} = \sum_{j \in \mathrm{occ}} \overline{v}_{ajij}.$$

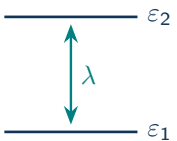
Canonical HF specialisation

In the usual HF-based partition, the residual reference-to-single coupling vanishes. Only the doubles term remains (Møller-Plesset second order, MP2). Changing the partition also changes the unperturbed energies and the residual operator.

A variant benchmark: purely off-diagonal coupling (section 9.5)

Now remove the diagonal interaction present in the earlier example:

$$H = H_0 + V, \qquad H_0 = \sum_{p=1}^{2} \varepsilon_p a_p^\dagger a_p, \qquad V = \lambda \sum_{p \neq q} a_p^\dagger a_q.$$



$$H = \begin{pmatrix} \varepsilon_1 & \lambda \\ \lambda & \varepsilon_2 \end{pmatrix}, \qquad \varepsilon_1 < \varepsilon_2,$$

$$V_{11} = V_{22} = 0, \quad V_{12} = V_{21} = \lambda.$$

One-particle basis: $|\Phi_1\rangle = a_1^\dagger|0\rangle$, $|\Phi_2\rangle = a_2^\dagger|0\rangle$.

The lower eigenvalue, with $\Delta = \varepsilon_2 - \varepsilon_1 > 0$:

$$\boxed{E_- = \frac{\varepsilon_1 + \varepsilon_2}{2} - \frac{1}{2}\sqrt{\Delta^2 + 4\lambda^2} = \varepsilon_1 + \frac{\Delta}{2}\left[1 - \sqrt{1 + \frac{4\lambda^2}{\Delta^2}}\right].}$$

Brillouin-Wigner: a two-vertex equation is exact here

Choose $P = |\Phi_1\rangle\langle\Phi_1|$ and $Q = |\Phi_2\rangle\langle\Phi_2|$. Because $V_{11} = V_{22} = 0$, all further Q-space interaction insertions vanish:

$$QVQ = 0, \qquad \boxed{E_- - \varepsilon_1 = \frac{\lambda^2}{E_- - \varepsilon_2},} \qquad \boxed{(E_- - \varepsilon_1)(E_- - \varepsilon_2) - \lambda^2 = 0,}$$

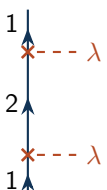
which is the exact secular equation.

An important qualification

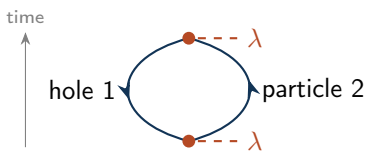
The energy-dependent equation is solved self-consistently. Its solution contains all even powers of λ, despite having only two explicit vertices. In this particular two-state model the BW solution is exact; this property does not hold for a generic two-vertex truncation (chapter 9: BW(2) and BW(3) for the pairing model are well off).

The same virtual excursion in two representations

$$1 \xrightarrow{\lambda} 2 \xrightarrow{\lambda} 1, \qquad \frac{V_{12}V_{21}}{E_- - \varepsilon_2} = \frac{\lambda^2}{E_- - \varepsilon_2}.$$



One-body transition chain.



Closed particle-hole representation.

Next week turns the right-hand picture into a rule: a cut between the two vertices gives the denominator, the loop gives the sign.

Rayleigh-Schrödinger terms through third order

Write $D = \varepsilon_1 - \varepsilon_2 < 0$. The denominators are now fixed:

$$E^{(1)} = V_{11} = 0, \qquad \boxed{E^{(2)} = \frac{V_{12}V_{21}}{D} = \frac{\lambda^2}{D}.} \qquad E^{(3)} = \frac{V_{12}V_{22}V_{21}}{D^2} - \frac{V_{11}V_{12}V_{21}}{D^2} = 0.$$

Therefore the total energy through third order is

$$\boxed{E_-^{[3]} = \varepsilon_1 + \frac{\lambda^2}{D}.}$$

The third-order coefficient vanishes; the sum through third order still contains the nonzero second-order term.

Fixed and self-consistent denominators differ at finite order

RS through third order

$$\Delta E^{[3]} = \frac{\lambda^2}{D}.$$

One fixed unperturbed energy denominator.

Self-consistent BW

$$\Delta E^{\mathrm{BW}} = \frac{\lambda^2}{D + \Delta E^{\mathrm{BW}}}.$$

The denominator contains the correlated energy shift.

Both start with λ^2/D, but the BW solution also contains higher even powers. Thus $\Delta E^{[3]} \neq \Delta E^{\mathrm{BW}}$ for finite nonzero λ. In this particular two-state model the BW solution is exact; this property does not hold for a generic two-vertex truncation.

The exact expansion identifies the fourth-order target

Using $D = \varepsilon_1 - \varepsilon_2 < 0$, expand the exact square root:

$$\boxed{E_- = \varepsilon_1 + \frac{\lambda^2}{D}\left[1 - \frac{\lambda^2}{D^2} + \cdots\right].} \qquad \boxed{E^{(4)} = -\frac{\lambda^4}{D^3} > 0.}$$

- The Taylor expansion about $\lambda = 0$ applies inside $|2\lambda/\Delta| < 1$.
- Low-order accuracy requires the coupling to be small relative to the gap.
- The exact square-root expression remains defined outside this expansion region.

Fourth-order RS terms in resolvent form

For this benchmark only, set

$$\langle X\rangle_0 = \langle\Phi_1|X|\Phi_1\rangle, \qquad R = \frac{Q}{\varepsilon_1 - H_0}, \qquad Q = |\Phi_2\rangle\langle\Phi_2|.$$

The fourth-order expression is

$$E^{(4)} = \langle VRVRVRV\rangle_0 - E^{(1)}\left[\langle VR^2VRV\rangle_0 + \langle VRVR^2V\rangle_0\right] + \left(E^{(1)}\right)^2\langle VR^3V\rangle_0 - E^{(2)}\langle VR^2V\rangle_0.$$

- $E^{(1)} = V_{11} = 0$ removes the middle terms.
- $QVQ = 0$ removes the first term.
- Only the final normalisation (folded) term survives.

The surviving fourth-order term

Since $R = |\Phi_2\rangle\langle\Phi_2|/D$,

$$E^{(2)} = \langle VRV\rangle_0 = \frac{\lambda^2}{D}, \qquad \langle VR^2V\rangle_0 = \frac{\lambda^2}{D^2}, \qquad \boxed{E^{(4)} = -E^{(2)}\langle VR^2V\rangle_0 = -\frac{\lambda^4}{D^3}, \qquad D < 0.}$$

Agreement with the exact lower eigenvalue

$$E_- = \varepsilon_1 + \frac{\lambda^2}{\varepsilon_1 - \varepsilon_2} - \frac{\lambda^4}{(\varepsilon_1 - \varepsilon_2)^3} + \mathcal{O}(\lambda^6).$$

The second-order shift is negative; the fourth-order correction is positive. A reader who drops the "normalisation corrections" gets the fourth order of this problem not approximately but completely wrong: there is nothing else.

Pause to think #5

(about 3 min – discuss with your neighbour)

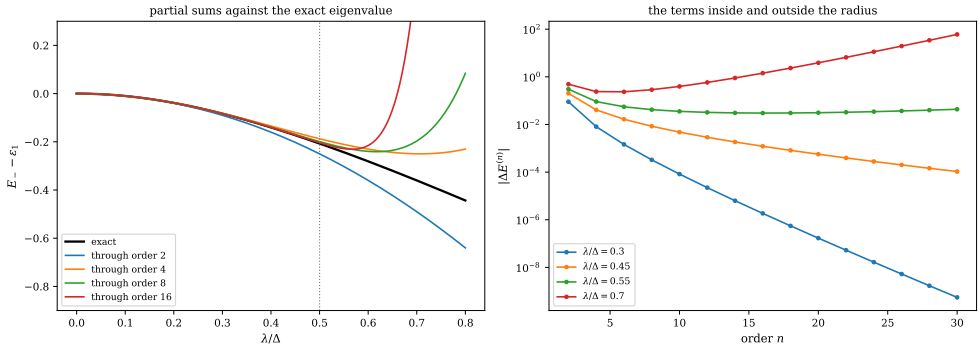
The exact eigenvalue $E_-(\lambda)$ is a smooth function of λ on the whole real axis, yet its Taylor series converges only for $|2\lambda/\Delta| < 1$.

1. Where are the singularities of $E_-(\lambda)$ that limit the radius? Are they ever visited by a real calculation?
2. At $\lambda/\Delta = 0.55$ the terms $|E^{(n)}|$ decrease for fifteen orders before they start to grow. What would a calculation stopped at fourth order conclude?

Answer to pause #5

1. At the branch points of the square root, $4\lambda^2 = -\Delta^2$, i.e. $\lambda = \pm i\Delta/2$: complex couplings, never visited, but they fix the radius of convergence about $\lambda = 0$ all the same.
2. That the series converges. Nothing in the first few orders announces the radius; the terms simply stop decreasing much later. Chapter 9 finds the same behaviour in the pairing model beyond $g \approx 1.5$, where no closed form says where the singularity sits.

The benchmark drawn: partial sums and terms



Left: exact E_- and RS partial sums through orders 2, 4, 8, 16 against λ/Δ; the dotted line is the radius $\lambda/\Delta = \frac{1}{2}$. Right: $|E^{(n)}|$ for $\lambda/\Delta = 0.3, 0.45, 0.55, 0.7$ – geometric decrease inside, eventual growth outside, from the sixth order at 0.7 and only from the sixteenth at 0.55. (`mbpt.py`, chapter 9, section 9.5.)

Friday, lecture 4: All contributions, listed

The recursion that generates every order

With $R_0 = Q/(\varepsilon_0 - H_0)$ and $|\Psi_0\rangle = \sum_n |\Psi^{(n)}\rangle$, $|\Psi^{(0)}\rangle = |\Phi_0\rangle$, collecting powers of η in $|\Psi_0\rangle = |\Phi_0\rangle + R_0(H_I - \Delta E)|\Psi_0\rangle$ gives

$$\boxed{|\Psi^{(n)}\rangle = R_0\Big[H_I|\Psi^{(n-1)}\rangle - \sum_{k=1}^{n-1} E^{(k)}|\Psi^{(n-k)}\rangle\Big], \qquad E^{(n+1)} = \langle\Phi_0|H_I|\Psi^{(n)}\rangle.}$$

- The subtracted terms are the normalisation ("renormalisation", "unlinked") corrections; they appear from $n = 2$ on.
- $R_0|\Phi_0\rangle = 0$: every $|\Psi^{(n)}\rangle$, $n \geq 1$, lies in Q space, so intermediate normalisation is preserved order by order.
- `mbpt.py` implements exactly this recursion and checks it against the explicit expressions on the next two frames.

All contributions to the wave operator through third order

Write $|\Psi_0\rangle = \Omega|\Phi_0\rangle$ with $\Omega = 1 + \Omega^{(1)} + \Omega^{(2)} + \Omega^{(3)} + \cdots$, $R \equiv R_0$, $V \equiv H_I$:

$$\Omega^{(1)} = RV,$$

$$\Omega^{(2)} = RVRV \; - \; E^{(1)} R^2 V,$$

$$\Omega^{(3)} = RVRVRV \; - \; E^{(1)}\left[RVR^2V + R^2VRV\right] \; + \; \left(E^{(1)}\right)^2 R^3 V \; - \; E^{(2)} R^2 V.$$

In the determinant basis, with $V_{mn} = \langle\Phi_m|H_I|\Phi_n\rangle$ and $D_m = \varepsilon_0 - \varepsilon_m$ $(m, n, I \neq 0)$:

$$\langle\Phi_m|\Psi^{(1)}\rangle = \frac{V_{m0}}{D_m}, \qquad \langle\Phi_m|\Psi^{(2)}\rangle = \sum_n \frac{V_{mn}V_{n0}}{D_m D_n} - \frac{V_{00}V_{m0}}{D_m^2},$$

$$\langle\Phi_m|\Psi^{(3)}\rangle = \sum_{nI} \frac{V_{mn}V_{nI}V_{I0}}{D_m D_n D_I} - V_{00}\Big[\sum_n \frac{V_{mn}V_{n0}}{D_m D_n^2} + \sum_n \frac{V_{mn}V_{n0}}{D_m^2 D_n}\Big] + V_{00}^2 \frac{V_{m0}}{D_m^3} - E^{(2)}\frac{V_{m0}}{D_m^2}.$$

For a two-body H_I: $\Psi^{(1)}$ has singles ($U_{ai}/(\varepsilon_i - \varepsilon_a)$) and doubles ($\overline{v}_{abij}/(\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b)$); $\Psi^{(2)}$ reaches up to quadruples; $\Psi^{(3)}$ up to sextuples. Rank is not order.

All contributions to the energy through fourth order

With $\langle X\rangle_0 = \langle\Phi_0|X|\Phi_0\rangle$, $R = Q/(\varepsilon_0 - H_0)$, $V = H_I$, and $E^{(n+1)} = \langle V\Omega^{(n)}\rangle_0$:

$$E^{(1)} = \langle V\rangle_0,$$

$$E^{(2)} = \langle VRV\rangle_0,$$

$$E^{(3)} = \langle VRVRV\rangle_0 \;-\; E^{(1)}\langle VR^2V\rangle_0,$$

$$E^{(4)} = \langle VRVRVRV\rangle_0 \;-\; E^{(1)}\big[\langle VR^2VRV\rangle_0 + \langle VRVR^2V\rangle_0\big] \;+\; \big(E^{(1)}\big)^2\langle VR^3V\rangle_0 \;-\; E^{(2)}\langle VR^2V\rangle_0.$$

In the determinant basis:

$$E^{(1)} = V_{00}, \qquad E^{(2)} = \sum_m \frac{V_{0m}V_{m0}}{D_m}, \qquad E^{(3)} = \sum_{mn}\frac{V_{0m}V_{mn}V_{n0}}{D_mD_n} - V_{00}\sum_m\frac{V_{0m}V_{m0}}{D_m^2},$$

$$E^{(4)} = \sum_{mnl}\frac{V_{0m}V_{mn}V_{nl}V_{l0}}{D_mD_nD_l} - 2V_{00}\sum_{mn}\frac{V_{0m}V_{mn}V_{n0}}{D_m^2D_n} + V_{00}^2\sum_m\frac{V_{0m}V_{m0}}{D_m^3} - \Big(\sum_n\frac{V_{0n}V_{n0}}{D_n}\Big)\sum_m\frac{V_{0m}V_{m0}}{D_m^2}.$$

(The two mixed terms are complex conjugates, equal for real matrix elements, hence the 2.) Every term with a factor $E^{(k)}$ in front is a renormalisation term: the product of two separately evaluated pieces. Next week: these are the *unlinked* diagrams, and they cancel against the unlinked parts of the principal term – the linked-diagram theorem.

The same list in the language of determinants, through second order

For $H_0 = \sum_p \varepsilon_p a_p^\dagger a_p$ and $H_I = E^{(1)} + U_N + V_N$:

$$E^{(1)} = \sum_i u_{ii} + \frac{1}{2} \sum_{ij \in \mathrm{occ}} \overline{v}_{ijij} \quad \text{(with } u = 0 \text{ for a diagonal } h_0\text{)},$$

$$E^{(2)} = \underbrace{\sum_{ia} \frac{|U_{ai}|^2}{\varepsilon_i - \varepsilon_a}}_{1p1h} + \underbrace{\frac{1}{4} \sum_{ijab} \frac{|\overline{v}_{abij}|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}}_{2p2h},$$

$$\Omega^{(1)} |\Phi_0\rangle = \sum_{ia} \frac{U_{ai}}{\varepsilon_i - \varepsilon_a} |\Phi_i^a\rangle + \frac{1}{4} \sum_{ijab} \frac{\overline{v}_{abij}}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b} |\Phi_{ij}^{ab}\rangle.$$

Measured

Four models, `mbpt.py`: the fourth order from the explicit formula and from the recursion agree to 10^{-12}. Hubbard ($U = 2$): the four pieces are -0.191, $+0.303$, -0.151, $+0.032$ and the total -0.0076. The cancellations are severe, and they are the point of next week.

Pause to think #6

(about 3 min – discuss with your neighbour)

Look at the list of fourth-order terms.

1. How many distinct terms does $E^{(4)}$ have for a reference with $E^{(1)} = 0$ (Hartree-Fock partition with the mean field in H_0)?
2. In the third order, $E^{(1)}\langle VR^2V\rangle_0$ is the product of two quantities that are each sums over the whole system. If the system were two non-interacting copies of a smaller one, how would this product scale with the number of copies? And $E^{(3)}$ itself?

Answer to pause #6

1. Two: the principal term $\langle VRVRVRV \rangle_0$ and the folded term $-E^{(2)}\langle VR^2V \rangle_0$. For the Lipkin model ($E^{(1)} = 0$) `mbpt.py` gives -0.0036 and $+0.0072$, total $+0.0036$.
2. $E^{(1)}$ and $\langle VR^2V \rangle_0$ each scale linearly with the number of copies, so their product scales *quadratically* – an energy cannot. The principal term $\langle VRVRV \rangle_0$ contains a quadratically scaling piece of its own, and the subtraction removes it exactly: $E^{(3)}$ scales linearly. This is size extensivity, verified to machine precision in chapter 9, and next week it becomes the statement that only *linked* diagrams contribute.

Summary

Summary of week 44

Thursday: the framework

- $E_0 - E_{\mathrm{ref}} = \sum c_i^a f_{ia} + \sum c_{ij}^{ab} \overline{v}_{ijab}$: singles and doubles couple directly
- Intermediate normalisation; $\Delta E = E_0 - \varepsilon_0$ versus $E_0 - E_{\mathrm{ref}}$
- P, Q, the reduced resolvent, an exact equation, its Neumann series
- BW: $\omega = E_0$, implicit, resummed into G_Q and $H_{\mathrm{eff}}(E_0)$, exact for the two-state model
- RS: $\omega = \varepsilon_0$, $E^{(1)} = \langle H_I \rangle_0$, $E^{(2)} = \langle H_I R_0 H_I \rangle_0$, $E^{(3)}$ with its normalisation subtraction

Friday: the terms

- $H_I = E^{(1)} + U_N + V_N$ with $U_{pq} = \sum_j \overline{v}_{pjqj}$: singles from U_N, doubles from V_N
- $E^{(2)} = \sum |U_{ai}|^2/(\varepsilon_i - \varepsilon_a) + \frac{1}{4} \sum |\overline{v}_{abij}|^2/(\ldots)$: coherent sum, four contractions, counting factor
- Two-level benchmark: BW exact with two vertices; RS $E^{(3)} = 0$, $E^{(4)} = -E^{(2)} \langle V R^2 V \rangle_0$; radius $|2\lambda/\Delta| = 1$
- The complete lists: $\Omega^{(1,2,3)}$ and $E^{(1,2,3,4)}$

The renormalisation terms are not corrections: they cancel exactly what would make the energy scale wrongly with system size.

Next week

Week 45, November 5–6: diagram rules and their derivation (chapter 9, sections 9.6–9.12)

Every term listed today is drawn: the perturbation as a cross (one-body remainder) and a dashed line (interaction), particle and hole lines, the reference as the vacuum, and the *energy denominator* as a cut between successive vertices. The rules – sums over labels, a denominator per cut, $\frac{1}{2}$ per pair of equivalent lines, the sign $(-1)^{h+l}$ – are derived from Wick's theorem, and the first-order, first-order-wave-function and second-order diagrams are read off and matched to today's formulas. Then third order, the unlinked diagrams, and the linked-diagram theorem: why the renormalisation terms exist and what they cancel. Numerics: the four models of chapter 9 order by order, size extensivity, and how far the series can be trusted.

Reading: chapter 9, sections 9.6–9.12; Shavitt and Bartlett, chapters 4–5. The week-44 exercises (next two frames, `Exercises/ExercisesWeek44.tex`) are discussed in the exercise session and answered on the week-45 slides.

Exercises, week 44

Exercise 1: the expansion from the projected equation

Let $H = H_0 + V$ and $|\phi_n\rangle$ be the eigenstates of H_0 and $|\psi_n\rangle$ the corresponding ones for H. Assume that the ground states $|\phi_0\rangle$ and $|\psi_0\rangle$ are not degenerate. Show that

$$E_0 - \varepsilon_0 = \frac{\langle\phi_0|V|\psi_0\rangle}{\langle\phi_0|\psi_0\rangle}, \qquad H|\psi_0\rangle = E_0|\psi_0\rangle, \qquad H_0|\phi_0\rangle = \varepsilon_0|\phi_0\rangle.$$

a) Define the operators $P = |\phi_0\rangle\langle\phi_0|$ and $Q = 1 - P$. Show that these operators are idempotent.
b) Show that for any z we have

$$|\psi_0\rangle = \langle\phi_0|\psi_0\rangle \sum_{n=0}^{\infty} \left(\frac{Q}{z - H_0}(z - E_0 + V)\right)^n |\phi_0\rangle,$$

$$E_0 = \varepsilon_0 + \sum_{n=0}^{\infty} \langle\phi_0|V \left(\frac{Q}{z - H_0}(z - E_0 + V)\right)^n |\phi_0\rangle.$$

c) Discuss these results for $z = E_0$ (Brillouin-Wigner perturbation theory) and $z = \varepsilon_0$ (Rayleigh-Schrödinger perturbation theory). Compare the first few terms in these expansions.

Note the overlap $\langle\phi_0|\psi_0\rangle$ in the denominator: this is the general form, before intermediate normalisation sets it to one.

Exercise 2: two particles in a single j-shell

Consider a system of two fermions with spin $s = 1/2$ in the pair-orbitals $|m_0\rangle$ and $|-m_0\rangle$ in a single shell $j = l + s$ with $2j + 1 > 2$, where l is the orbital momentum with $l = 0, 1, 2, \ldots$. Assume that the matrix elements for the interaction between the particles take the form

$$\langle m, -m|v|m', -m'\rangle = -G.$$

a) Show that the Brillouin-Wigner expansion from the previous exercise can be used to give

$$E_0 = -(j + 1/2)G.$$

b) Show thereafter by direct diagonalisation of the Hamiltonian matrix that this is the exact energy. Use thereafter Rayleigh-Schrödinger perturbation theory and discuss the differences.

Many-body connection

Hint for the discussion in b): all $j + 1/2$ pair states are degenerate in H_0. Which of the two schemes survives that, and why? (Today's reduced resolvent is the place to look.)